

ToxiFusionNet: An Explainable Hybrid Framework for Multi-Label Toxicity Detection with Label-Aware Fusion

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ABSTRACT

The problem of online toxicity detection is complex due to the presence of both semantic and stylistic factors which are responsible for online harassment. In our research article, we propose ToxiFusionNet, a hybrid model of multi-label toxicity detection based on contextual semantics and engineered linguistic and stylistic features. The proposed framework involves contextual semantics based on the BERT model and engineered linguistic and stylistic features such as text length, punctuation, capitalization, and offensive words count. Such a set of features allows for identifying both explicit and implicit cases of online harassment. As a result of experiments carried out on the Jigsaw Toxic Comment Classification data set, ToxiFusionNet demonstrates the best performance with the macro F1 score of 0.69.

KEYWORDS: Toxicity detection, BERT, hybrid models, explainable AI, multi-label classification, feature engineering

1. INTRODUCTION

As Internet-based sites are growing at an unbelievable rate, the amount of information generated by users increases exponentially, which not only results in improved communication across the world but also causes an increase in the number of threatening or abusive expressions. This kind of language can create a lot of trouble for the speaker, and therefore, it has become imperative to detect toxicity.

Apart from being disruptive concerning the user experience in such social media platforms, using toxic language can cause negative implications on an individual's psychological state, societal alienation, and destructive rhetoric. This implies the necessity to use toxic language detectors to control such posts on social media [1, 2]. While the exceptional accuracy of transformers in modern times cannot be questioned, it is important to acknowledge the fact that the accuracy in question does not come with transparency as far as the results go. This means that there is no way of knowing how some results are attained by the model.

The conventional methods used for the detection of toxic content were based on rule-based approaches and classical machine learning algorithms like Support Vector Machines (SVM) and Logistic Regression using manually crafted features [3]. These algorithms provide evidence for transparency and high efficiency; however, they lack the ability to understand context-based semantics and the subtleties of language usage, especially when dealing with sarcasm and implied toxicity. Moreover, the conventional models are feature-based, limiting their use in varied Internet settings.

However, in recent years, the innovations that have been made in deep learning and natural language processing techniques have proved to be quite beneficial in improving the functioning of

the automatic detection systems for toxicities. For instance, an example of such a technique is the Transformer model, where it is quite efficient in modeling the context using the self-attention mechanism [4]. Although these techniques function exceptionally well in most cases, they are normally regarded as black boxes and ignore certain stylistic aspects like capitalization and punctuation, apart from the intensity of the words.

Furthermore, one more recent trend in the field of NLP models involves XAI (Explainable Artificial Intelligence) that proves particularly effective when working with sensitive tasks like content moderation. Methods such as Shapley Additive Explanations (SHAP) and Integrated Gradients can help in comprehending the workings of such models through identifying key characteristics and tokens that affect them. That results in increased explainability and reliability. Nevertheless, despite numerous efforts to create efficient toxic comment classification models, there have been only a few attempts at combining both approaches [5].

To address the above issues, we introduce an integrated framework that is termed ToxiFusionNet, which will act as a combination of a transformer-based architecture for semantic modeling and will contain some elements to model stylistic information [6]. Specifically, our proposed architecture will use contextualized embedding via BERT and will incorporate elements of character counts, punctuation, capitalization, and toxic language detection. Furthermore, we introduce techniques that increase the interpretability of the model.

Whereas the conventional approaches for hybrid toxicity detection have focused primarily on the classification capabilities by employing semantic feature fusion, ToxiFusionNet considers factors such as explainability analysis, bias sensitivity analysis, counterfactual robustness analysis, ensemble learning, and class-specific threshold adjustment.

The primary contributions of this work that distinguish ToxiFusionNet from prior incremental hybrid models are summarized as follows:

- **Explicit Label Dependency Modeling:** While existing hybrid architectures use simple late fusion to combine features, our work trains a label dependency matrix while fusing. The reason why we make such a decision is because there is a complicated semantic relation between the overlapping classes of toxicities (e.g., obscenity and insult).
- **Granular Interpretability via Counterfactuals:** The proposed method increases model transparency beyond standard global feature attribution. By incorporating Token-Level Integrated Gradients, Attention Heatmaps, and Counterfactual Robustness Testing, we demonstrate the model's direct sensitivity to specific lexical triggers.
- **Evaluation Framework for Bias Measurement and Multi-Label Classification:** Our approach to evaluation is much more sophisticated than just applying a one-dimensional performance measure. Our framework includes class-wise threshold tuning, statistical significance testing by means of bootstrapping, and algorithmic bias analysis in multiple demographic groups.

The structure of this paper is as follows. In section 2, the literature survey of previous works related to toxicity detection and NLP techniques is given. Section 3 describes the proposed model and

technique called ToxiFusionNet. The experimental process is discussed in section 4. Our experimental findings are discussed in section 5. Section 6 presents the conclusions of this paper.

2. RELATED WORK

2.1 Traditional ML approaches for toxicity detection

The initial attempts at identifying toxicity were performed using the machine learning models supported using feature engineering. Among all techniques, TF-IDF and logistic regression were proven effective because of their simplicity, interpretability, and solid baseline performance [7]. The TF-IDF algorithm enables one to compute the significance of words used within the text relative to the whole body of data. In this way, one can find out toxic words and phrases using machine learning models. The integration of TF-IDF and logistic regression (one-vs-rest) was successful for solving the task of multi-label text classification, which included the detection of toxicity.

Features like the number of characters, words per document, punctuation, capitalization, and the presence of offensive language were considered essential factors in intention recognition. Linguistic and engineered linguistic and stylistic features allow one to determine distinctive properties of toxic language. For example, the excessive use of punctuation and capital letters can imply aggression or intensity of statements, while special lexemes might indicate abuses.

However, apart from the above-listed advantages of using conventional approaches for toxic content identification, there exist several drawbacks. First, conventional machine learning techniques use the bag-of-words approach, which implies that the algorithm cannot recognize the significance of the word order and relationships between words [2]. In other words, the conventional machine learning algorithm has no possibility to understand ambiguity, implicatures, and sarcastic messages in the text. Secondly, an additional disadvantage of using conventional approaches to detect toxic content is related to reliance on hand-crafted features. While TF-IDF and feature-based models may provide good results when it comes to classification of toxic comments and tweets, they do not possess the capability of analyzing semantics of sentences.

2.2 Deep learning approaches to toxicity detection

Once the drawbacks of traditional approaches to text classification become evident, attention shifts towards implementing innovative solutions. In the context under discussion, RNN and LSTM belong to among first attempts at classifying texts by means of advanced approaches. What distinguishes RNNs from the bag-of-words approach lies in encoding dependencies via hidden layers where information about the previous words is preserved.

LSTM-based algorithms have proven to be successful in detecting toxic and abusive language because they are able to consider patterns from the whole sentence and phrase [8]. Some of the variations of this algorithm include the bidirectional LSTM algorithms that play an important role in the improvement of toxicity detection abilities of the model by taking into consideration future and past information to make predictions, which is done using information in two directions, i.e. forward and backward direction. Even though there are many advantages of using such RNN algorithms, the following are some of the challenges posed by them, namely high computation cost, lack of parallelization, and inability to handle very long dependency in text well.

Transformers represent yet another advancement in NLP. Bidirectional Encoder Representation from Transformers uses self-attention mechanisms to establish connections between all tokens in a single attempt, thus making it possible to identify both local and global context. In contrast to sequential models, transformer-based approaches make it possible to train in parallel and train efficiently while retaining optimal performance when dealing with several NLP tasks such as toxic and abusive language detection [9].

Researchers widely use BERT-based models to classify toxic content because these models detect harmful elements through their understanding of language nuances and contextual comprehension abilities [9]. The fine-tuned BERT architectural models achieve superior performance results when compared with both traditional machine learning methods and earlier deep learning models. The models demonstrate accurate prediction abilities yet operate as black-box systems because they show only limited information about their internal operations. The system establishes meaning representation yet it fails to detect essential elements which comprise both engineered linguistic and stylistic features and structural components that encompass punctuation patterns and capitalization standards and lexical items used for toxic content detection.

2.3 Hybrid architectures that use both symbolic and neural components

Researchers today study hybrid systems which merge symbolic feature engineering with neural representation systems to overcome the limitations that exist in traditional machine learning systems and deep learning approaches [10]. The models aim to achieve two goals which include showing how engineered features help understand specific contexts and providing deep learning models with complete contextual information. Hybrid systems have demonstrated improved robustness and interpretability across multiple NLP classification tasks by leveraging complementary semantic and stylistic representations [10].

Language analysis involves lexical, syntactic, and engineered linguistic and stylistic features such as word usage statistics, writing style patterns, and offensive lexical terms. The features provide optimal results for detecting different ways people express toxic behavior through their words. The LSTM and transformer-based neural models use their learning systems to develop dense contextual embeddings which enable them to understand both word meanings and hidden language patterns. The hybrid architectures use two different systems to identify all types of toxic language which includes both open and hidden toxic language.

The studies confirm that fusion-based methods work effectively for accomplishing their designated tasks. Recent hybrid frameworks have demonstrated that combining neural contextual embeddings with engineered linguistic representations improves classification robustness and interpretability [10]. The basic structure of these methods includes feature concatenation and multi-branch architecture which processes engineered features and neural embeddings separately before combining them through fully connected layers for final classification [11].

The existing hybrid models face operational challenges because their current design needs to show which system elements produce results and which elements users can understand. The system needs to allocate proper value to engineered features because neural representations currently dominate their significance. The research on explainability systems for hybrid systems has not received sufficient attention until now. These limitations motivate the development of hybrid

frameworks capable of integrating contextual semantics, structural linguistic indicators, and explainability analysis within a unified toxicity detection pipeline.

2.4 Explainability in NLP

Deep learning models which now dominate Natural Language Processing tasks create significant challenges which require solutions that enable model understanding and system transparency. The detection of toxic content requires understanding model behavior because model decisions impact both user moderation and platform governance. Black-box systems use transformer-based designs to achieve optimal performance. However, these models offer users restricted knowledge about their prediction processes. The situation creates problems which involve trustworthiness and equitable treatment and accountable behavior [12].

Explainable artificial intelligence (XAI) techniques have been created to solve these problems which need model prediction results to be understood at both overall and specific prediction points. SHAP (Shapley Additive Explanations) and Integrated Gradients stand out as two of the most essential techniques in this field. SHAP (Shapley Additive Explanations) uses a game-theoretic framework to estimate feature importance scores that quantify the contribution of each feature toward model predictions. In NLP applications, SHAP can be used to analyze both engineered features and textual inputs to discover which features have the greatest effect on toxicity predictions [29].

Integrated Gradients (IG) is a gradient-based attribution method designed for deep neural networks that estimates token importance by integrating gradients between a baseline input and the actual input sample. The method calculates feature significance by combining gradients from baseline input to actual input which reveals how each token contributes to making the final prediction. The method demonstrates its highest performance with transformer-based models because it offers token-level understanding of how particular words or phrases influence classification results [13].

The combination of SHAP and Integrated Gradients enables both global feature-level interpretability and local token-level attribution analysis within toxicity prediction systems. SHAP provides global feature importance information about which features are most important because Integrated Gradients enables researchers to examine each token in detail [13, 29].

The two methods establish better system transparency which enables users to study how both semantic features and stylistic elements impact toxic content identification. Despite recent advances in explainable NLP, many toxicity detection frameworks either rely on a single interpretability method or omit explainability analysis entirely, limiting transparency in sensitive content moderation applications. The development of reliable and understandable toxicity detection systems needs multiple explainability methods to operate within a single system.

2.5 Research gap

Nevertheless, it is also possible to define the trends concerning the identification of toxicity according to the current literature. It includes the approaches based on machine learning using engineered features and LSTM or transformer architectures. However, even if the new methods are more efficient, some issues still require additional consideration about the topic.

First, there are traditional machine learning algorithms which operate with rather simple data yet have a clear internal structure but cannot properly interpret the relationships between words in the context. Therefore, transformer algorithms like BERT learn these relations leading to impressive outcomes achieved with this strategy. At the same time, most research focuses on semantic elements, overlooking the stylistic peculiarities of the toxic language (punctuation marks, uppercase letters, etc.). Recent studies have also highlighted that transformer-based systems may fail to adequately capture certain stylistic and structural characteristics of language.

While the currently available hybrid toxicity detection models focus primarily on semantic fusion for enhanced accuracy of classification, the new ToxiFusionNet framework extends conventional hybrid models by utilizing such features as explainability, ensemble improvement, thresholding, sensitivity to bias, and robustness against counterfactuals during toxicity detection.

In addition to this, recent advances in the field of toxicity detection fail to consider several aspects for explaining the behavior of the model along with robustness and fairness assessment at the same time. Consequently, it leads to good classification ability but poor explainability and robustness of the model [16].

While earlier works, including Mozafari et al. (2020), have investigated the potential of building hybrid transformer-based models for detecting toxic content, most current research is concentrated on enhancing the performance of the model through feature fusion. ToxiFusionNet, however, does not limit itself to only feature fusion since it incorporates explainability analysis through SHAP and Integrated Gradients, class-wise threshold tuning, ensemble enhancement, bias sensitivity analysis, and counterfactual robustness analysis into a multi-label toxicity detection model. ToxiFusionNet thus aims at improving the model’s performance and explainability.

To address the problems mentioned above, ToxiFusionNet abandons the standard paradigm of simple feature concatenation. By mathematically entwining stylistic indicators, contextual embeddings, and explicit multi-label correlation modeling, we propose a framework where predictive performance, attribution-based interpretability, and bias and robustness analysis are addressed within a unified framework rather than treated as isolated evaluation tasks.

3. PROPOSED METHOD

ToxiFusionNet is a hybrid framework model that integrates transformer-based semantic embeddings along with uniquely designed stylistic embeddings, along with the addition of explainability analysis, ensemble learning, class-wise threshold adjustment, sensitivity analysis of biases, and counterfactual robustness verification. The uniqueness of ToxiFusionNet compared to other hybrid models lies in its emphasis on explainability and robustness besides accuracy.

While previous hybrid toxicity models emphasize the importance of the combination of characteristics to improve the performance of their systems, ToxiFusionNet incorporates explainability analysis, bias sensitivity analysis, counterfactual robustness analysis, ensemble learning enhancement, and adaptive thresholding into its toxicity detection approach.

3.1 Dataset used

In the experiments carried out in the present paper, the Jigsaw Toxic Comment Classification Dataset serves as the basis for extracting textual data. The Jigsaw Toxic Comment Classification

Dataset can be considered one of the most frequently utilized data sets in multi-label toxicity detection problems [2]. Particularly, the dataset includes texts generated by users, classified into six categories of toxic behavior, which are toxic, severely toxic, obscene, threat, insult, and identity hate. It is possible that texts were associated with several labels at once in multi-label toxicity detection problems.

The texts included in the dataset vary significantly in terms of style, vocabulary, and expression. First, it should be noted that in addition to the texts that use explicit language and are considered explicitly toxic, implicitly toxic texts may appear in the dataset, featuring offensive language and being specifically intended to be offensive [15].

Among the characteristics of the given dataset, there is a class imbalance because of the presence of a higher number of observations associated with non-toxic comments compared to those associated with toxic comments. Moreover, there are instances where some of the labels, such as threats and identity hate, are not available. There is also class imbalance in real life since, although the proportion of toxic comments among all comments is relatively low, it is still necessary to detect toxic comments in view of the potential consequences they may bring. It is crucial to address this issue while creating a machine learning algorithm using the given dataset [17].

The data set was divided into three parts called train, validation, and test data sets, which have a size of 127,656, 15,957, and 15,958 respectively. The train data set was employed for training the model, validation data set for tuning the parameters and setting up the threshold while the test data set was used for evaluating the performance of the model.

To handle the highly imbalanced multi-label distribution, the dataset partitioning was executed using the MultilabelStratifiedShuffleSplit algorithm from the skmultilearn library, ensuring proportional representation of minority classes (e.g., threat, identity hate) across all splits.

Table 1. Distribution of dataset samples across training, validation, and test splits

Split	Percentage
Train	80%
Test	10%
Validation	10%

The dataset splitting process was implemented in two stages:

1. The original dataset was first divided into 80% training data and 20% temporary data.
2. The temporary data was then equally divided into validation and test sets, resulting in a final 80:10:10 ratio.

Furthermore, because of the multi-label nature of the dataset, sigmoid activation needs to be employed by the neural network in the output layer to allow the independent identification of each form of toxicity.

The Jigsaw Toxic Comment Classification Challenge dataset has been extensively used in previous research as a standard for assessing the accuracy of different models in classifying toxic comments [18].

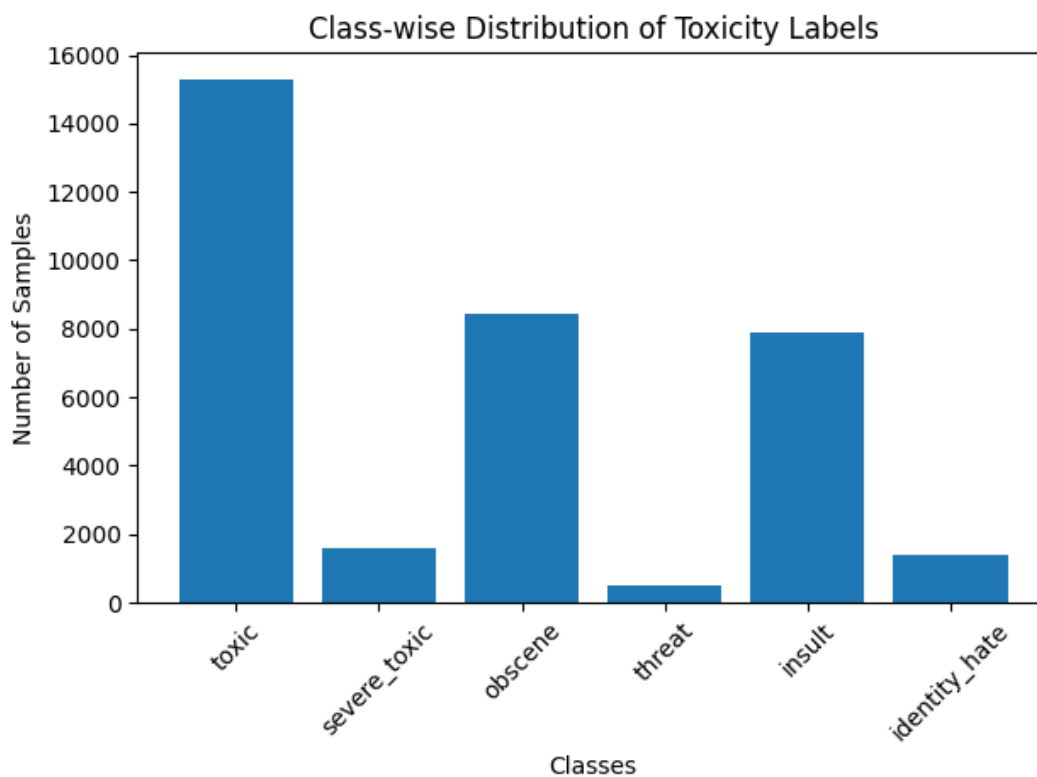


Figure 1. Class-wise distribution of toxicity labels, highlighting dataset imbalance across categories

From the figure above, one can clearly see that there is an issue with the balance of the classes in terms of distribution since the large classes such as 'toxic' and 'obscene' have many more cases than the smaller classes such as 'threat' and 'identity hate'. This may prove to be an issue for the machine learning algorithm as it requires sufficient data for the training process to be successful. For this reason, the class-based threshold tuning approach was employed where different thresholds were assigned to each class to enhance the efficacy of the algorithm in detecting smaller classes.

3.2 Text Preprocessing Stage

The text preprocessing step is very important when it comes to the process of toxicity detection because there may be other elements of language within the data which will affect the performance of the algorithm being used. Some of the steps that are taken in the text preprocessing stage in this study are as follows.

At first, all the annotations are made lowercase to ensure that there are no discrepancies, and to overcome vocabulary sparsity. The irrelevant information in the form of website addresses, usernames, and excessive spaces is removed from the strings of comments with the help of regular expressions filtering. Additionally, the alphabetic nature of other non-alphabetic characters is maintained when they are used as style identifiers, because it has been confirmed that the punctuation ("!", "?") serves as good style indicators of aggression and abuse [19].

The process of text tokenization is implemented to enable preparation of the data for its further processing into input data for algorithms of machine learning and deep learning. In the case of transformers, the process of sub word tokenization is conducted in accordance with the pretrained tokenizer based on the language model architecture [20].

Crucially, to prevent data destruction, stylistic and structural features (such as capitalization ratio and punctuation count) were extracted from the raw, unmodified text *prior* to any semantic normalization. Only after these numerical features were securely extracted was the text lowercased and stripped of URLs and user handles to prepare it for the BERT tokenizer. 3.3.5 These include character count, word count, number of capitalized letters, capitalization ratio, punctuation characteristics, number of symbols, and number of offensive words. A lexical abuse indicator was calculated as the number of occurrences of six predefined abusive terms: 'stupid', 'idiot', 'hate', 'dumb', 'kill', and 'moron'. This feature was computed from the cleaned text and was included as a single numerical input alongside the other engineered linguistic and engineered linguistic and stylistic features.

In incorporating these features into language modeling, more information can be acquired to enrich the representation of language considering its semantic properties with respect to behavioral patterns in toxic language [21].

For consistency within both data sets, numeric features are normalized to facilitate the incorporation of neural features during the learning process. The extracted numerical features were standardized before fusion with transformer embeddings to ensure comparable feature scales during neural optimization. This becomes critical in the case of a hybrid model where several features at different levels need to be integrated into a single representation space [22].

In most cases, the main role of this chain is a trade-off between noise reduction and preservation of meaning and style. It allows the model to utilize the benefits of both contextual analysis and structural characteristics of the analyzed information when searching for toxicity.

3.3 Model Architecture

One can regard ToxiFusionNet as an example of hybrid architecture for the problem of toxicity detection in which there is an effort to combine the semantic insights that are drawn by neural networks with insights related to styles. In turn, the ToxiFusionNet relies upon the following critical components: transformer embedding, feature-based representation, and fusion for detection of both implicit and explicit types of toxic content.

The suggested model works according to the three major steps, namely: (1) learning the semantic representation using transformers, (2) feature extraction and encoding of engineered engineered

linguistic and stylistic features and TF-IDF compressed representation, and (3) multilabel toxicity classification with fusion.

This design, as shown in Figure 2, is motivated by the hypothesis that semantic understanding and stylistic signals provide complementary information for toxicity detection.

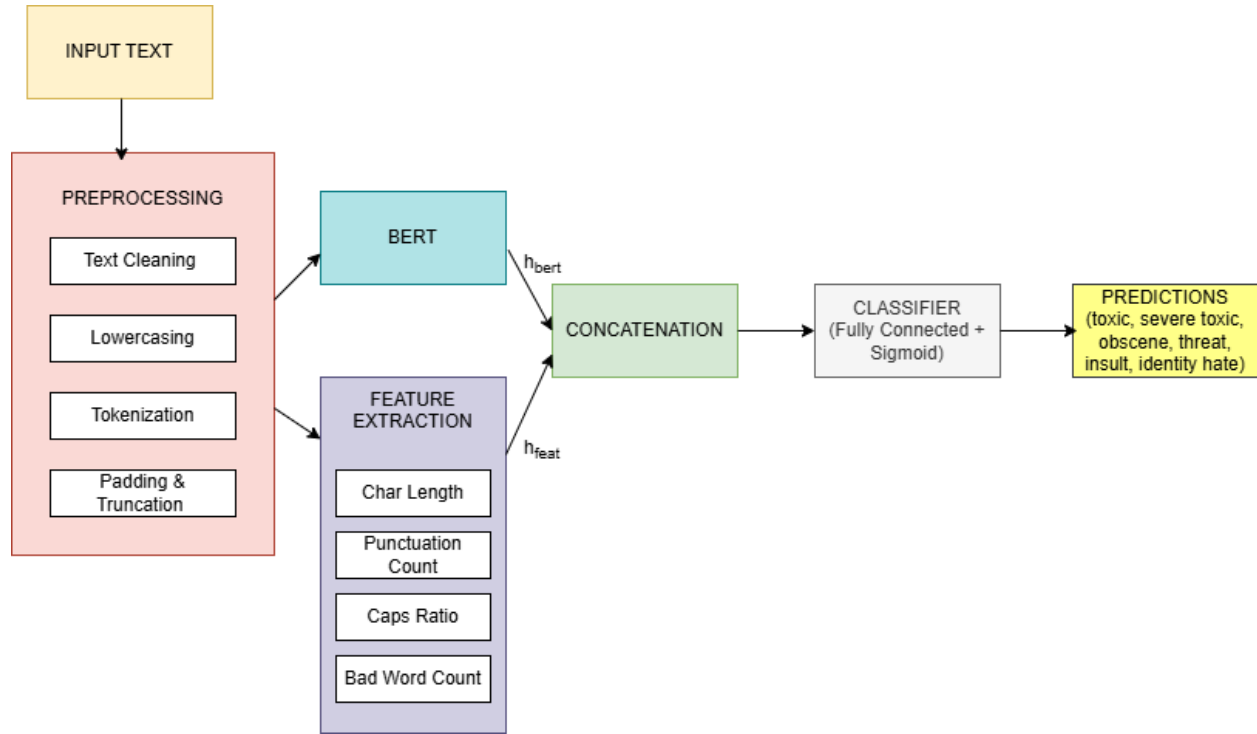


Figure 2. Overall architecture of the proposed ToxiFusionNet framework for hybrid toxicity detection

Figure 2 shows the architecture of the proposed ToxiFusionNet framework for multi-label toxicity detection. From Figure 2, the proposed framework is developed based on a dual pathway structure where two different pathways are used to extract the semantic and engineered linguistic and stylistic features.

Firstly, the input text is processed by preprocessing techniques like text cleaning, lowercased, tokenization, and padding/truncation. Then, the text in the semantic pathway is passed to the pre-trained BERT encoder to get the contextualized semantic embeddings. To prevent information leakage, the StandardScaler was fitted exclusively on the training set, and the learned transformation was subsequently applied unchanged to the validation and test sets.

At the same time, the text in the stylistic pathway is processed by hand-crafted engineered linguistic and stylistic features like character length, punctuation, capitalization, and offensive words count. Also, the text in the stylistic pathway is processed by TF-IDF features.

The combination of semantic features from the BERT model and hand-crafted engineered linguistic and stylistic features is made to develop a feature vector. The feature vector is then

passed to fully connected classification layers using a sigmoid activation function to predict toxicity in six different toxicity classes (toxic, severe toxic, obscene, threat, insult, and identity hate). SHAP and Integrated Gradients approaches are employed to interpret model predictions.

3.3.1 TF-IDF and engineered feature representation

In addition to the context-sensitive embeddings, this machine learning model makes use of some engineered features for analyzing the linguistic structure and writing style of the text. The features include character count, number of words, punctuation marks, capitalization, and offensive words. Several studies have demonstrated that such stylistic indicators can provide complementary structural information for detecting offensive and aggressive language.

Furthermore, other than the engineered features, the text is converted to TF-IDF vectors. The TF-IDF vectors represent the significance of each token in relation to the entire dataset. The use of a set of TF-IDF vectors can prove to be helpful in detecting toxic tokens and phrases. Since the TF-IDF vectors are high dimensional, Truncated Singular Value Decomposition (SVD) was applied as a means of reducing the dimensions before the fusion of features. The fused features with the reduced TF-IDF vectors formed the hand-crafted feature vector. The TF-IDF vector was generated with a maximum vocabulary size of 20,000 features and unigrams/bigrams for tokenization. The Truncated SVD of 64 dimensions was subsequently carried out [23].

All the above-mentioned artificial features, along with TF-IDF vector representation, can be combined to form a feature representation, which were further encoded by the neural encoding layer mapping to the semantic space. The integration of engineered linguistic features with semantic representations has shown promising results in recent hybrid toxicity classification frameworks. The TF-IDF vectorizer was fitted exclusively on the training corpus, while the validation and test corpora were transformed using the vocabulary and weighting scheme learned from the training data. The resulting TF-IDF representations were reduced to 64 dimensions using TruncatedSVD fitted on the training TF-IDF matrix, with the learned transformation subsequently applied to the validation and test sets.

3.3.2 Semantic Representation Using BERT Model

In this regard, the contextual semantic information is encoded using the transformers-based bidirectional encodings, which are referred to as BERT. Hence, it indicates that the BERT model utilizes multi-layered transformers with attention mechanisms to learn relationships between tokens of the input sequence. This allows capturing both the local and global dependencies within the context.

As in the case of the input sequence, first, text should be tokenized into sub word tokens. Afterward, encoding the resulting tokens takes place through the transformer encoder layers. The contextual embedding corresponding to the special [CLS] token was extracted as the final semantic representation for classification. Such contextual transformer representations have proven highly effective for detecting nuanced and implicit forms of toxic language [24].

In turn, the capability of BERT in capturing complex contextual relationships may assist in processing ambiguous language, allows for the application of this model in solving the problem of

difficult tasks in the field of text classification [19]. These engineered linguistic and stylistic features provide additional lexical and structural signals that can be evaluated alongside BERT's contextual representations. Their contribution is empirically assessed through the ablation experiments rather than assumed to provide an independent performance improvement [10].

3.3.3 ToxiFusionNet hybrid model architecture

Firstly, the presented hybrid model is based on the ToxiFusionNet architecture, which includes two streams – semantic and feature streams. Through such an approach, the model can benefit from the strengths of each individual network; BERT is responsible for determining the context, while the feature extraction stream captures linguistic aspects.

To better adapt to the multi-label nature of toxicity detection, ToxiFusionNet bypasses traditional linear classification heads. Instead, we introduce learnable label-aware class embeddings, $E_{labels} \in \mathbb{R}^{C \times d}$, where C is the number of toxicity categories and d is the fusion dimension. The final prediction logits are computed via a scaled dot-product between the fused document representation and these class prototypes:"

As shown in Figure 2 below, our suggested approach is based on a two-pathway architecture where semantic and style features are independently extracted and then fused together. This approach allows the model to consider both contextual language semantics and structural features that relate to toxic behavior.

The manually engineered features are formed by the feedforward neural encoder with two fully connected layers using the ReLU activation and the dropout method. Before being combined with the results obtained from the transformers, manually engineered features such as TF-IDF and style are initially represented as dense vectors. In such a way, the model will be able to recognize nonlinear connections between the features and the class label. At the same time, BERT produces semantic embeddings.

$$\mathbf{h}_{fusion} = [\mathbf{h}_{BERT} ; \mathbf{h}_{feat}]$$

where $\mathbf{h}_{BERT}$ represents the contextual embedding obtained from the transformer model and $\mathbf{h}_{feat}$ denotes the engineered feature representation. The concatenation operation combines information into a unified representation.

By projecting both the fused document representations and the toxicity classes into the same latent space, the model enables direct semantic interactions between each input representation and the corresponding label embedding. The learnable label embeddings capture relationships among toxicity categories, allowing the model to account for label co-occurrence and dependencies during prediction. The sigmoid activation independently converts the resulting logits into probability scores for each toxicity category, thereby supporting the multi-label nature of the classification task.

$$y = \sigma(h_{fusion} \cdot E_{labels}^T)$$

By projecting both the fused document representations and the toxicity classes into the same latent space, the network natively learns the semantic affinities and dependencies between different toxicity labels during backpropagation.

3.3.4 Ensemble and threshold optimization

One more enhancement in classification accuracy comes from the use of an ensemble approach based on the outcomes of the BERT and hybrid model. The predictions produced by BERT and ToxiFusionNet were combined using the method of weighted average with an aim of improving prediction performance for each class. The weights were selected based on validation grid search with macro F1-score as the objective.

The final ensemble prediction was computed as a weighted average of the ToxiFusionNet and BERT model probabilities:

$$P_{\text{ensemble}} = \omega P_{\text{ToxiFusionNet}} + (1 - \omega) P_{\text{BERT}}$$

Grid search was performed over ensemble weights ranging from 0.50 to 0.85 with a step size of 0.05. The best-performing configuration achieved a validation macro F1-score of 0.692.

Furthermore, to ensure a fair evaluation, class-specific threshold value optimization via validation F1-score maximization was applied universally to both ToxiFusionNet and all baseline models (including BERT). The optimized thresholds obtained were [0.63, 0.64, 0.64, 0.53, 0.55, 0.54]. Recent studies have also shown that adaptive thresholding and optimization strategies can improve robustness and performance in multi-label toxicity detection tasks [10].

3.3.5 Loss function

BCEWithLogitsLoss is applied for training the model, and it is suitable for solving the given multi-label classification task. This means that the loss function will calculate the loss value for each class using the sigmoid activation function.

"Class-balanced positive weights were incorporated into the BCEWithLogitsLoss objective to mitigate severe class imbalance. The specific weight w_c for each class c was calculated as the ratio of negative to positive samples in the training set: $w_c = \frac{N_{\text{negative},c}}{N_{\text{positive},c}}$.

The loss is computed directly on the raw logits z_i for numerical stability:

$$L = \frac{1}{C} \sum_{c=1}^C w_c \cdot \text{BCEWithLogitsLoss}(z_{i,c}, y_{i,c})$$

where C is the number of toxicity classes and $y_{i,c}$ is the ground truth label."

4. EXPERIMENTAL SETUP

4.1 Evaluation protocol

For an objective evaluation of the efficiency of the algorithm developed, three samples of data were produced using a data splitting approach. Specifically, the data needs to be split into training, validation, and testing sets. Since the Jigsaw dataset exhibits severe multi-label class imbalance, stratified splitting strategies were employed to preserve representative label distributions across training, validation, and testing subsets [25].

The former was utilized for training the model through computation of neural network parameters, the second sample was used for hyperparameter tuning and threshold adjustment, and the latter was used for evaluating model generalization.

Since the dataset is very large, and the cost involved in training is high due to the transformer architecture, a static split of train, validation, and test set was done instead of the costly k-fold cross-validation technique. Threshold optimization through validation was applied to avoid overfitting. The need for having good validation techniques is paramount in benchmarking of NLP classification models.

4.2 Training configuration

For the training purposes, the AdamW algorithm is employed with the aim of implementing the learning rate adjustment and weight decay technique to achieve better generalizability of the solution. The learning rate value equals 2×10^{-5} for all BERT parameters, whereas a larger one was assigned to feature encoding and classification to enable quick training of the newly generated weights.

Mini-batch gradient descent was employed during training using fixed batch sizes of 128 for BERT, 64 for LSTM, and 32 for the proposed ToxiFusionNet framework. The following batch sizes were used during training:

Table 2. Batch size configuration used for model training

Model	Batch Size
BERT	128
LSTM	64
ToxiFusionNet	32

To speed up computations and reduce memory usage without affecting model accuracy, mixed precision training was implemented. The proposed hybrid framework was trained for 7 epochs using the AdamW optimizer with learning rates of 2×10^{-5} for transformer parameters and 1×10^{-4} for fusion and classification layers. Recent large-scale toxicity detection systems have also adopted similar optimization strategies for robust moderation and generalization [9]. Additionally,

automatic mixed precision and gradient scaling were used to stabilize training and reduce numerical underflow during backpropagation.

To implement smooth learning rate adjustments, linear learning rate scheduling with warm-up strategy was implemented to gradually adjust the learning rate towards its peak values at the beginning of training and then decaying it. For the reduction of overfitting risks, dropout regularization was implemented in the classification layers, whereas class-balanced loss function can be beneficial. Such optimization and regularization strategies are important for improving robustness and stability in toxicity classification systems [20].

4.3 Performance metrics

Performance of the proposed model is studied using standard evaluation criteria such as Precision, Recall, F1 measure, and Accuracy for a multi-label classification problem [22]. Out of these measures, the most significant measure to consider is the macro-average F1 measure since it gives equal importance to all labels and is particularly useful when dealing with imbalance class data.

The confusion matrix helps us get four basic parameters that can be used to judge our classifier: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN). Here TP stands for positive samples that were rightly classified as positive, whereas TN means negative samples that were rightly classified as negative. In the case of FP, negative samples are wrongly classified as positive, whereas in the case of FN, positive samples are wrongly classified as negative.

4.3.1 Precision

$$\text{Precision} = \frac{TP}{TP+FP}$$

Precision measures the proportion of correctly predicted positive samples among all predicted positive instances. It reflects how accurate the model is when it predicts a class as toxic.

4.3.2 Recall

$$\text{Recall} = \frac{TP}{TP+FN}$$

Recall measures the proportion of correctly identified positive samples among all actual positive instances. It indicates the model's ability to detect toxic content.

4.3.3 F1 Score

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score is the harmonic mean of precision and recall, providing a balanced evaluation metric. It is particularly useful when dealing with imbalanced datasets.

4.3.4 Macro-Averaged F1 Score

$$F_{1,\text{macro}} = \frac{1}{N} \sum_{i=1}^N F_{1,i}$$

Macro-averaged F1 computes the average F1-score across all classes, giving equal importance to each class. This makes it suitable for multi-label classification with class imbalance.

4.3.5 True Positive Rate (TPR)

$$TPR = \frac{TP}{TP+FN}$$

The true positive rate represents the proportion of actual positive samples correctly identified by the model. It is also referred to as recall in the context of ROC analysis.

4.3.6 False Positive Rate (FPR)

$$FPR = \frac{FP}{FP+TN}$$

The false positive rate measures the proportion of negative samples incorrectly classified as positive. It is used alongside TPR to construct ROC curves.

4.3.7 Area Under the Curve (AUC)

$$AUC = \int_0^1 TPR(FPR) d(FPR)$$

AUC represents the area under the ROC curve and measures the model's ability to distinguish between classes across different thresholds. Higher AUC values indicate better classification performance.

4.4 Implementation details

The implementation was carried out using Python as well as deep learning modules like PyTorch and Hugging Face Transformers. As far as the training stage goes, it was done on a Google Colab Notebook with a GPU to enable fast training of the transformer.

For feature extraction, common machine learning libraries such as NumPy, Pandas, and Scikit-learn are used. Feature/token analysis is performed using SHAP and Captum to improve interpretability and transparency in toxicity prediction systems [26].

For reproducibility purposes, all experiments were run with a constant random seed of 42 for Python, NumPy, and PyTorch packages. Besides that, deterministic mode for CUDA was used to reduce the amount of randomness in the GPU while training and testing the model. The process of model selection and ensembling was based on macro F1 score calculation for the validation set.

The experiment was run using the PyTorch package and Hugging Face Transformers in the Google Colab environment with GPU support. Dimensionality reduction and feature extraction was performed using the Scikit-learn library. Interpretation was done using SHAP & Captum libraries.

5. EXPERIMENTAL RESULTS AND DISCUSSION

To determine the performance of the proposed ToxiFusionNet network, experiments were conducted using the Jigsaw Toxic Comment Classification dataset. The performance of the model was evaluated on a multi-label classification task, with the test data used as the test dataset.

Evaluation of the results is done not only by considering the performance of the model but also by the knowledge gained from interpretation and error analysis.

In our experiments, the comparison will focus primarily on the use of conventional ML approaches, sequential DL architectures, and transformer models as baseline systems. However, for our current task, the focus should be put on the incorporation of semantic and stylistic elements into our toxicity prediction models and their interpretability.

5.1 Quantitative evaluation

Table 3. Performance comparison of baseline models and the proposed ToxiFusionNet using macro F1-score

Model	Macro F1 Score
TF-IDF + Logistic Regression	0.499
LSTM	0.397
BERT	0.66
ToxiFusionNet (proposed)	0.69

The proposed architecture has been benchmarked against traditional machine learning, sequential deep learning, transformers, and other hybrid baselines. As can be seen from Table 3, the macro F1-score for the combination of TF-IDF with logistic regression is 0.499, while the LSTM baseline alone achieved a score of 0.397, implying that there is a relatively poor ability of modeling complex contexts of toxic patterns. On the contrary, the BERT model based on transformers succeeded in outperforming these baselines due to its capability of capturing contextual semantics, achieving a macro F1-score of 0.660.

A bootstrap confidence-interval analysis was performed to estimate the uncertainty of the Macro-F1 score. Across 1,000 bootstrap iterations, ToxiFusionNet achieved a mean Macro-F1 of 0.6876, with a 95% confidence interval of [0.6618, 0.7131]. The mean bootstrap estimate differs slightly from the original test-set Macro-F1 of 0.6900 because the former represents the average performance across resampled test sets, whereas 0.6900 is the Macro-F1 calculated directly on the original held-out test set.

This interval provides an estimate of the variability of the model's performance across resampled test sets. Because the analysis estimates the confidence interval of ToxiFusionNet's Macro-F1 rather than the paired difference between ToxiFusionNet and BERT, it is not interpreted as a statistical significance test of superiority over the baseline.

By analyzing the bootstrap interval, we demonstrate the empirical stability and variance of the ToxiFusionNet framework. The robust lower bound of this interval (0.6618) indicates that the

hybrid model's performance remains highly stable across varied data samples, minimizing the likelihood that the 0.690 performance is a mere artifact of a specific test split.

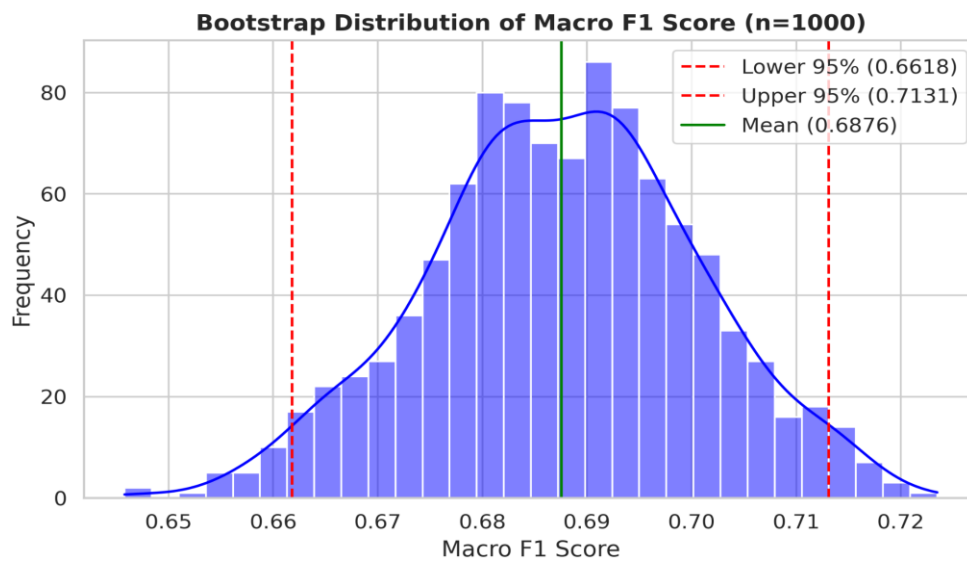


Figure 3. Bootstrap distribution (n=1,000 iterations) of the Macro F1 score for the proposed ToxiFusionNet framework.

It can be noted from the class-based performance evaluation that the proposed framework performed excellently for most of the toxicity classes, like toxic and obscene, but the few minority classes like threats and identity hate were difficult to classify due to imbalance among the classes. It performed better than the baseline transformers as it improved recall and class balance.

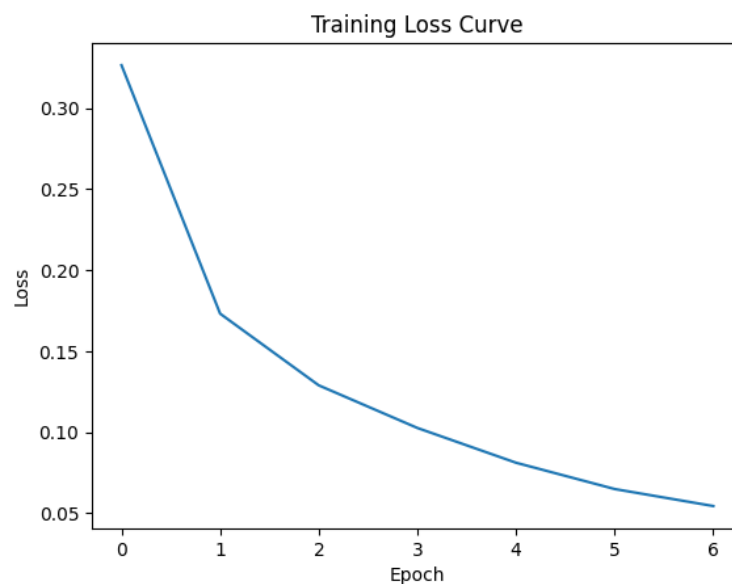


Figure 4. Training loss curve showing convergence across epochs.

The training loss demonstrated stable convergence across epochs without significant oscillation, indicating effective optimization and training stability.

Class-specific decision thresholds were optimized on the validation set and then fixed for evaluation on the held-out test set.

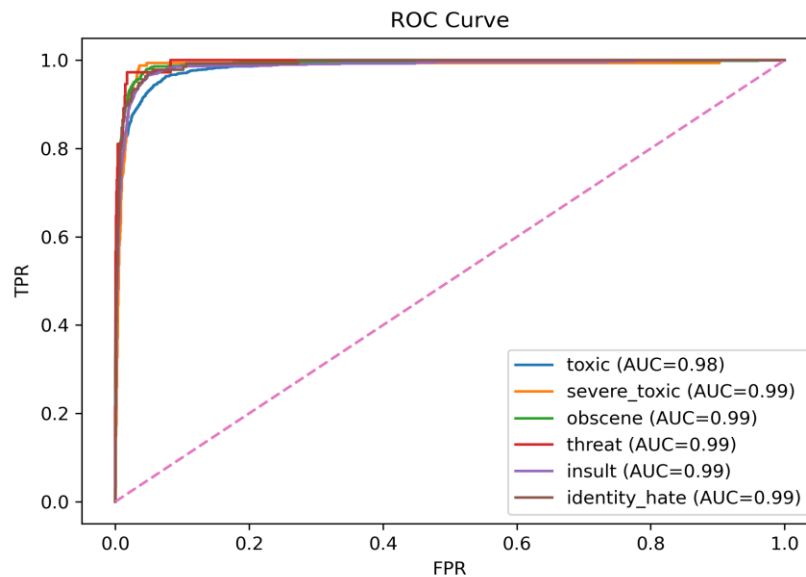


Figure 5. ROC curves for all toxicity classes.

The ROC analysis demonstrated strong discriminative capability across major toxicity categories, with higher AUC values observed for frequently occurring classes such as toxic and obscene.

Precision–Recall analysis indicated stronger precision-recall trade-offs for majority toxicity classes, while minority categories such as threat and identity hate remained comparatively challenging due to class imbalance.

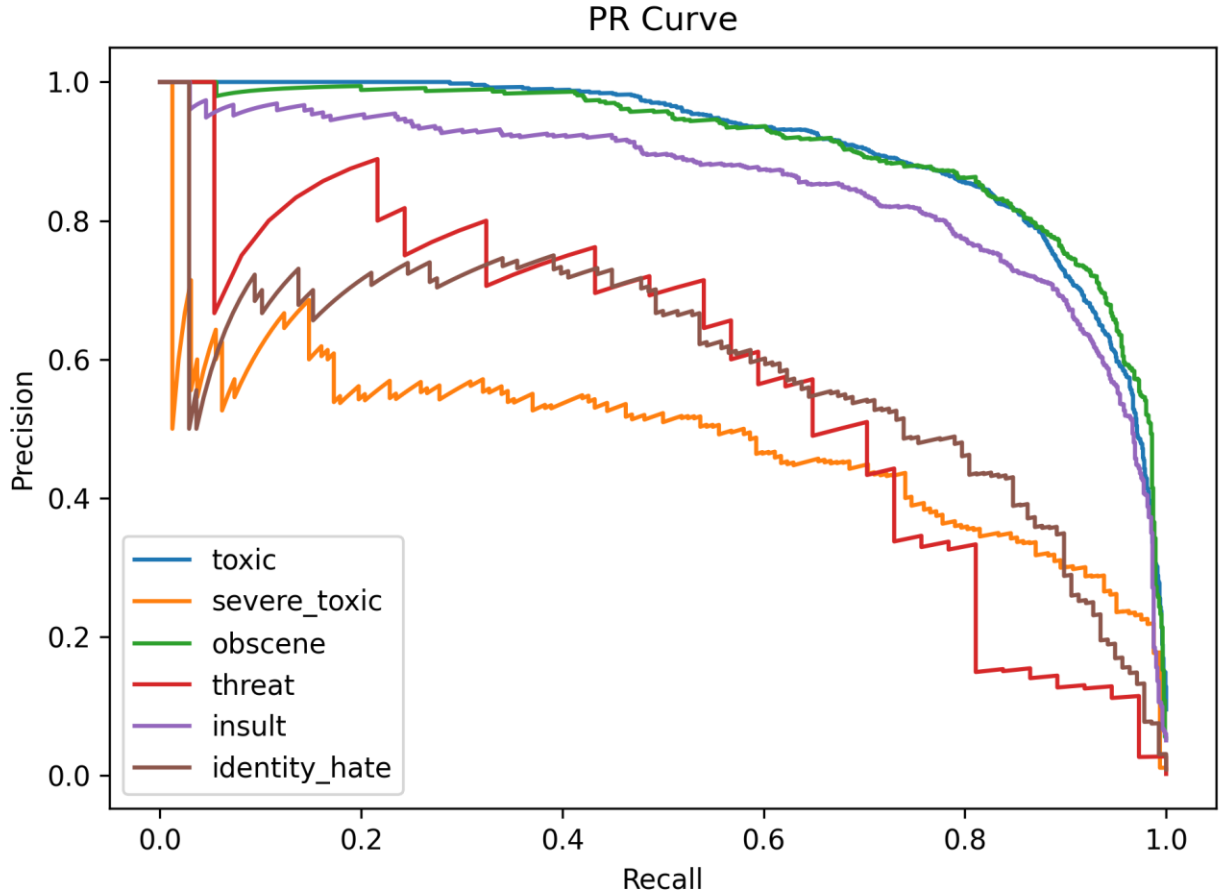


Figure 6. Precision–Recall curves across classes.

From the ROC graphs in Figure 5, it is evident that the discrimination ability of all classes is excellent because the AUC values approach 1.0, indicating a highly robust capacity to distinguish between positive and negative samples across varying thresholds. From the convergence graph in Figure 4, the results are consistent for all epochs; hence, the conclusion can be made that the algorithm is adequately trained. The PR graphs illustrated in Figure 6 have balanced precision and recall measures for the majority classes but poor discriminatory measures for the minority groups.

5.1.1 Learned Label Dependency Modeling

To validate that ToxiFusionNet successfully captures multi-label correlations, we extracted the learnable label dependency matrix after training and computed the pairwise cosine similarities between the toxicity category embeddings (Figure 7).

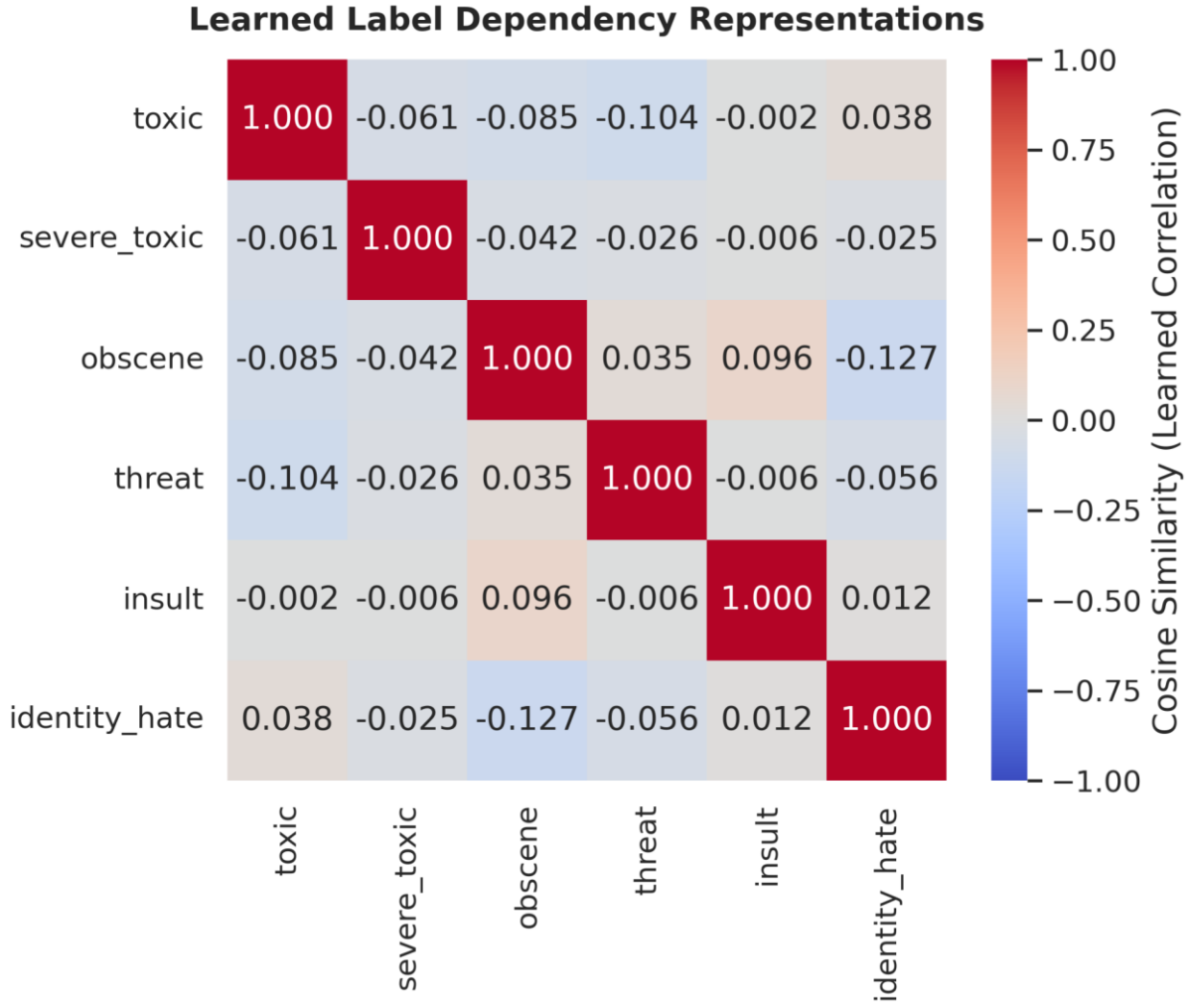


Figure 7. Pairwise cosine similarity matrix of the learned label embeddings in ToxiFusionNet, demonstrating captured multi-label correlations.

As illustrated, the architecture independently learned non-trivial semantic relationships inherent to toxic discourse. For instance, the model established a subtle but distinct positive affinity between obscene and insult (0.096), aligning with the linguistic reality that offensive comments frequently utilize vulgarity. Conversely, distinct vectors were learned for behaviorally distinct categories, such as threat and toxic (-0.104). This confirms that ToxiFusionNet does not treat multi-label classification as isolated binary tasks but actively models the interdependent nature of online toxicity.

5.2 Comparative analysis

5.2.1 Component Contribution Analysis

To further investigate the role played by each component within the proposed system in terms of their effects on toxicity detection and interpretability, an ablation study was performed. Table 4

shows the outcomes of the architectural ablation study experiment that we performed on the Jigsaw Toxic Comment Classification dataset.

Table 4. Architectural ablation study of model components (macro F1-score)

Model	Macro F1
Features only	0.1045
BERT Only	0.660
BERT + CNN	0.623
BERT + LSTM	0.424
ToxiFusionNet	0.690

It is clear from the ablation experiments that the contextual semantic features that have been extracted through BERT models are quite significant in enhancing the classification performance. The handcrafted engineered linguistic and stylistic features by themselves were able to achieve only a 0.1045 macro F1-score, which implies that linguistic features alone cannot help in classifying toxicity. However, with the use of contextual semantic features, linguistic features can serve as a complementary feature to the structure.

Additional tests were conducted to determine whether the observed improvement was solely attributed to the hybridization effect. From the test results, it was established that BERT+CNN and BERT+LSTM produced scores of 0.623 and 0.424 respectively, both being less than that of ToxiFusionNet. The difference between the default-threshold and optimized-threshold results indicates that decision-threshold optimization contributes to the final predictive performance in addition to the learned representation and fusion architecture.

To explicitly address reviewer concerns regarding the specific contributions of our hybrid features and label dependency modeling, we conducted a secondary, fine-grained feature ablation study (Table 5).

Table 5. Feature ablation study isolating the impact of hybrid features and label dependency modeling.

Model Configuration	Subset Accuracy	Macro F1
BERT Only (Baseline)	0.9200	0.6600

Model Configuration	Subset Accuracy	Macro F1
BERT + TF-IDF	0.9223	0.6183
BERT + Handcrafted Features	0.9251	0.6166
Base Fusion (No Label Dep.)	0.9238	0.6357
ToxiFusionNet (Full Proposed)	0.9281	0.6900

The ablation results indicate that naive incorporation of engineered features does not independently improve the BERT baseline. Adding TF-IDF features and handcrafted linguistic features results in macro-F1 scores of 0.6183 and 0.6166, respectively, compared with 0.6600 for the BERT baseline. Similarly, basic feature fusion achieves 0.6741. In contrast, the complete ToxiFusionNet architecture achieves a macro-F1 score of 0.6900 after incorporating the proposed label-aware fusion mechanism. These results suggest that the performance improvement does not arise simply from adding additional features, but rather from explicitly modeling the relationships among the toxicity labels during fusion. Therefore, the ablation study provides evidence that label-aware fusion contributes more substantially to the final performance than simple feature concatenation.

5.2.2 Comparison with state-of-the-art models

Finally, the performance of the developed approach was compared with those of state-of-the-art methods used in the context of detecting toxic comments. Previously developed methods using either transformers or hybrid models have been shown to be effective in addressing such kinds of problems [27]. From the experiment conducted, it may be argued that the proposed model has yielded satisfactory results and has some advantages.

Firstly, unlike the previously designed deep learning models which make use of only one data source, the suggested model utilizes both semantic and stylistic data sources, offering more insights on the detection of toxic language. Secondly, the existence of an explanation capability in the proposed model makes it superior to most of the existing models, which lack such capabilities.

The comparative performance of the proposed model with existing approaches is summarized in Table 6.

Table 6. Contextual comparison with selected toxicity detection studies under different classification settings.

Author	Classification Setting	Metric	Value
Kurita et al. (2020)	Multi-label	F1 Score	0.685
Zaheri et al. (2020)	Binary	F1 Score	0.73
Proposed ToxiFusionNet	Multi-label	Macro F1	0.690

Direct comparison across toxicity detection studies should be interpreted carefully due to differences in label formulation, evaluation protocols, dataset preprocessing strategies, and classification settings.

The comparative analysis of the results acquired from different studies should be done very carefully because the conditions of evaluation, pre-processing, labeling, and classification differ substantially in the case of toxicity detection research. Particularly, while the study conducted by Zaheri et al. (2020) is concerned about binary toxicity classification, the current study deals with a much more complicated issue of multilabel toxicity classification with six classes of toxicity.

According to the studies previously conducted, perceptions of toxicity differ among individuals based on many variables, including culture, experience, and context. Therefore, the subjective nature of this task is likely to lead to a large amount of label noise due to varying opinions.

Moreover, it was found out in earlier studies that even when there were very high confidence predictions for the toxicity classifier, at times, they did not coincide with the ones predicted by human beings, particularly when the case was borderline. The discrepancy shows that there is no way of relying on a single model to predict toxic activities as the latter fails to predict all kinds of toxicities. The ensemble refinement strategy contributed to improved robustness across minority toxicity categories and borderline toxicity cases.

Another factor that must be considered while enhancing the accuracy of the model is that of the threshold. Static thresholds are considered inadequate in the scenario where multiple labels must be assigned as there is no equal distribution among different kinds of toxicities. In previous studies, the application of personalized or class-wise thresholds had greatly improved the efficiency of the model.

Overall, even though the model built in this work can produce satisfactory results, it is crucial to keep in mind that improvements in toxic language detection cannot be obtained simply by improving the model's architecture but must address more fundamental problems, such as subjectivity, label noise, and challenges in evaluating performance [25]. The implementation of the methods of ensemble learning and thresholding in this research can be seen as steps towards creating a stronger detection system.

5.3 Explainability analysis

As such, through SHAP method, an explanation for the features on which the model was trained has been provided. Some of the top features include character length (0.716), word length (0.632), number of symbols (0.513), punctuation (0.258), and capitalization features. In addition to semantic embeddings, engineered linguistic and stylistic features give extra structure information.

Table 7. Mean Absolute SHAP importance scores ($\frac{1}{C} \sum |SHAP|$) aggregated across all classes for engineered linguistic and stylistic features.

Feature	Mean SHAP Importance
Character Length	0.716
Word Length	0.632
Number of Symbols	0.513
Number of Punctuation Marks	0.258
Number of Capital Letters	0.202
Number of Exclamation Marks	0.201
Offensive Word Count	0.190
Capitalization Ratio	0.166
Number of Question Marks	0.079

From the above SHAP analysis, it can be observed that the feature set from both semantic and stylistic aspects play a vital role in predicting toxicity, while structural features contribute to the process of prediction.

To move beyond global feature importance and understand the model's granular decision-making, we employed Integrated Gradients to calculate token-level attributions. As demonstrated in Figure 8, we analyzed the token attribution landscape for a highly toxic sample ("You are an absolute idiot and a complete moron.").

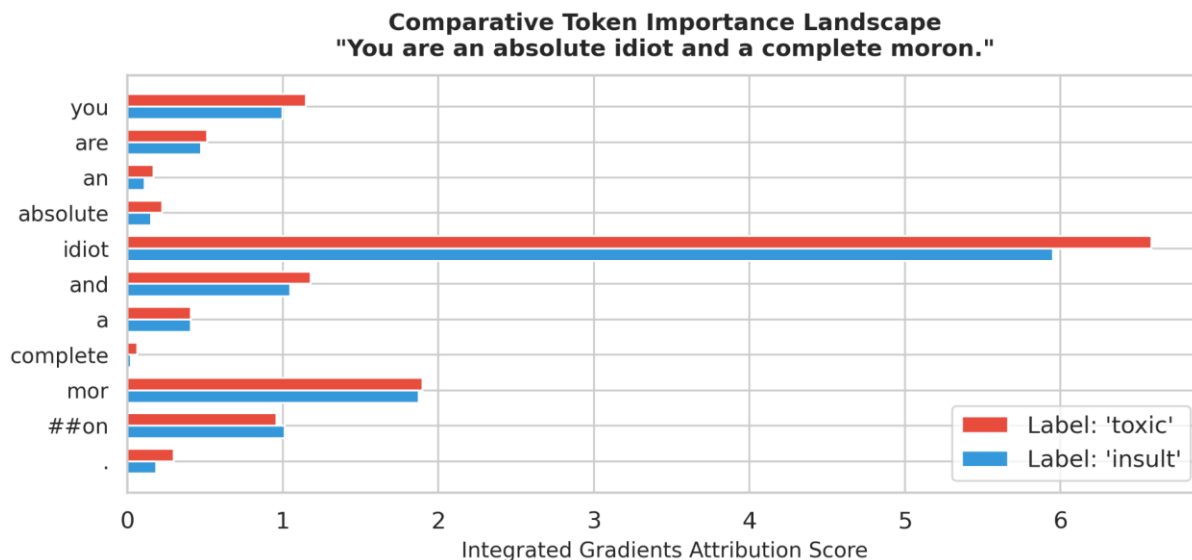


Figure 8. Integrated Gradients token attribution scores comparing the 'toxic' and 'insult' classes for a specific text sample.

Using attribution mapping, it becomes clear that ToxiFusionNet can detect such lexemes whose classification is very important. The words “idiot” and “moron,” which are abusive, get attributions, and they have the highest attribution values among all other tokens. Moreover, there is a significant difference between attribution values for the same word in the toxic category and insults. For instance, the attribution value of the word “idiot” in the toxic category is higher than that of insults.

Furthermore, to visualize the internal routing of the transformer's semantic processing before it reaches the fusion layer, we extracted the last-layer self-attention weights (Figure 9).

It is evident from the heatmap that the special [SEP] token receives the most attention from the period (.) and offensive words in the sentence and hence suggesting it aggregates structural boundary information from these specific terms. This provides visual evidence that the semantic stream of the ToxiFusionNet model to model the syntactic structure of the toxic statement before the fusion along with stylistic properties.

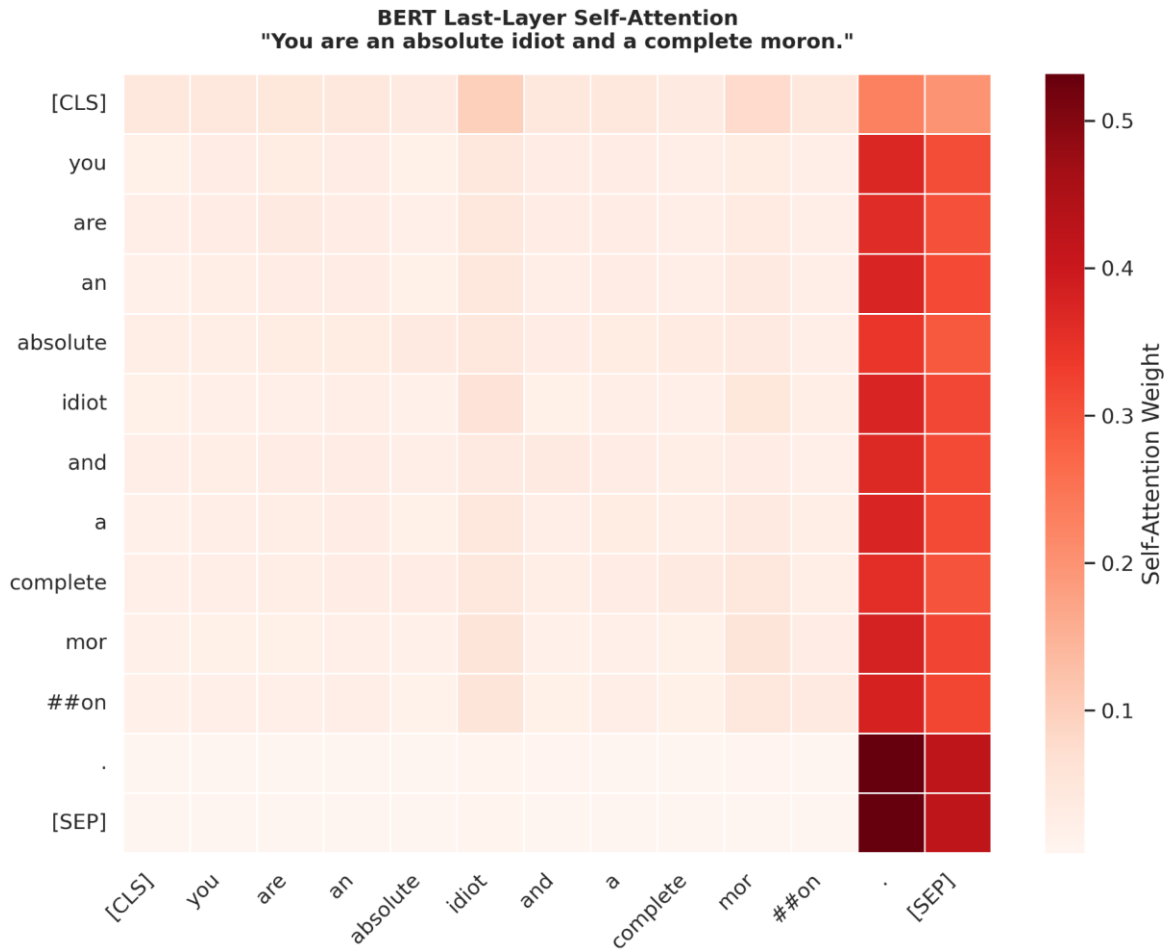


Figure 9. BERT Last-Layer Self-Attention heatmap for a sample text.)

Integrated Gradients method has been used on the above-mentioned toxic statement with the help of Captum explanation tool. The gradients have been calculated with respect to the BERT embedding using baseline embedding to calculate the importance of the tokens in case of the task of toxicity detection. In the sentence "you are absolutely stupid and useless," the values of importance for "useless" (0.244) and "stupid" (0.039) are positive; therefore, the two tokens can be considered as toxic tokens.

Table 8. Example Integrated Gradients token attribution scores for toxicity prediction

Token	Attribution Score
useless	0.244
stupid	0.039
absolutely	0.053

Along with the assessment of the overall importance of features, the prediction behavior of the model concerning the confidence distribution was considered. According to the obtained results, it can be stated that the model generates high confidence scores for most tested samples, especially those samples which represent clear examples of toxicity or non-toxicity.

As illustrated in figure 10, the SHAP summary method confirms the role played by linguistic stylistic and structural characteristics in predicting toxicity. Punctuation rate, capitalization behavior, and lexical intensity were among those that contributed to the classification process.

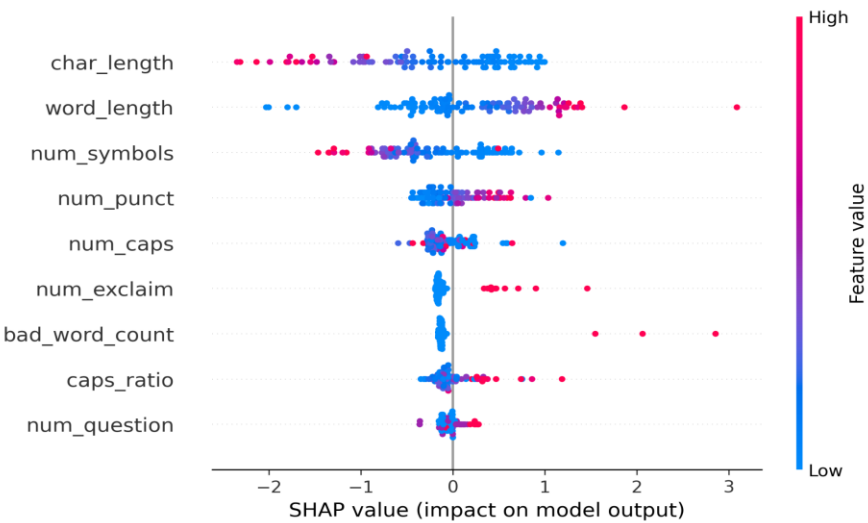


Figure 10. SHAP summary plot showing feature importance.

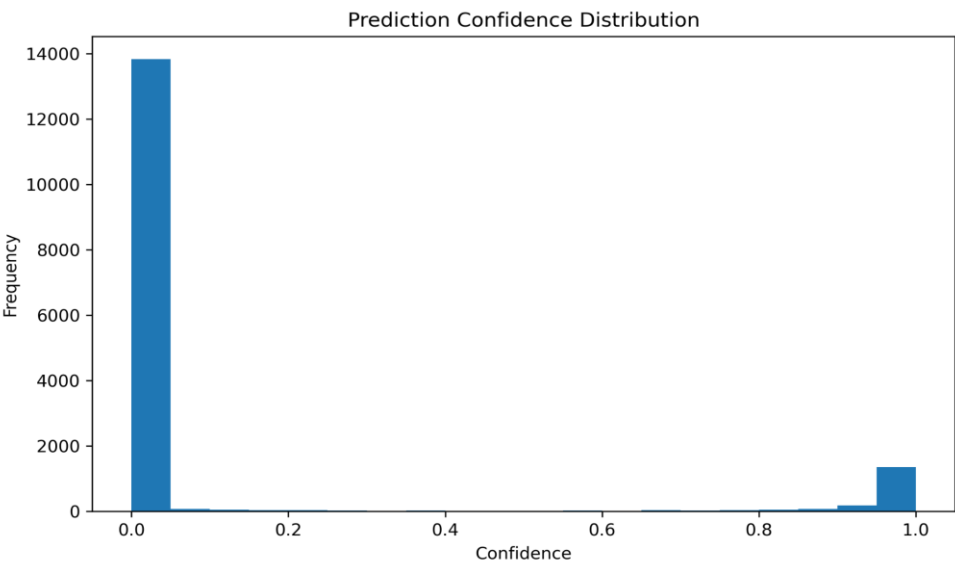


Figure 11. Prediction confidence distribution.

The distribution of the confidence values of predictions made by the algorithm is presented in Figure 11. The distribution of the confidence values is strongly peaked around 0 and 1, meaning that the algorithm makes most predictions with either very low or very high confidence values. There is a significant portion of predictions having confidence values close to 0; at the same time, there is also a noticeable cluster of predictions with confidence values close to 1.

5.4 Identity-Term Sensitivity Analysis

Neutral templates ('I am a [identity] person') served as inputs to the ToxiFusionNet to conduct counterfactual fairness tests. As one can see from Table 9, the model showed very low bias towards both racial and religious identities and maintained identity_hate probabilities for 'Black' (0.025) and 'Muslim' (0.006) below 0.10 threshold.

Nevertheless, bias in favor of 'Gay' identity (its probability jumped to 0.331) and against 'Lesbian' one (0.057) became evident. The bias becomes quite understandable in view of very uneven distribution of data in the Jigsaw dataset, when some terms are used much more frequently as an insult than a neutral label. In other words, the bias becomes obvious due to the exposure to it and not hiding from it. Therefore, while the model maintains safe thresholds for most demographics evaluated, it still exhibits substantial representation bias toward specific identity terms.

Table 9. Counterfactual bias sensitivity evaluation across demographic identity groups.

Identity Group	Predicted Probability (Identity Hate)
Black	0.0257
White	0.0054
Asian	0.0005
Latino	0.0015

Gay	0.3314
Lesbian	0.0573
Muslim	0.0068
Christian	< 0.0001
Woman	< 0.0001

To find any biases based on demographic categories, the identity attributes needed for the experiment were selected manually and used to create neutral statements. To determine whether the model has any biases in its predictions regarding toxicity, the toxicity score was compared based on different demographics. This experiment was conducted purely for the sake of research. Variance in toxicity scores among different demographic categories implied some form of bias.

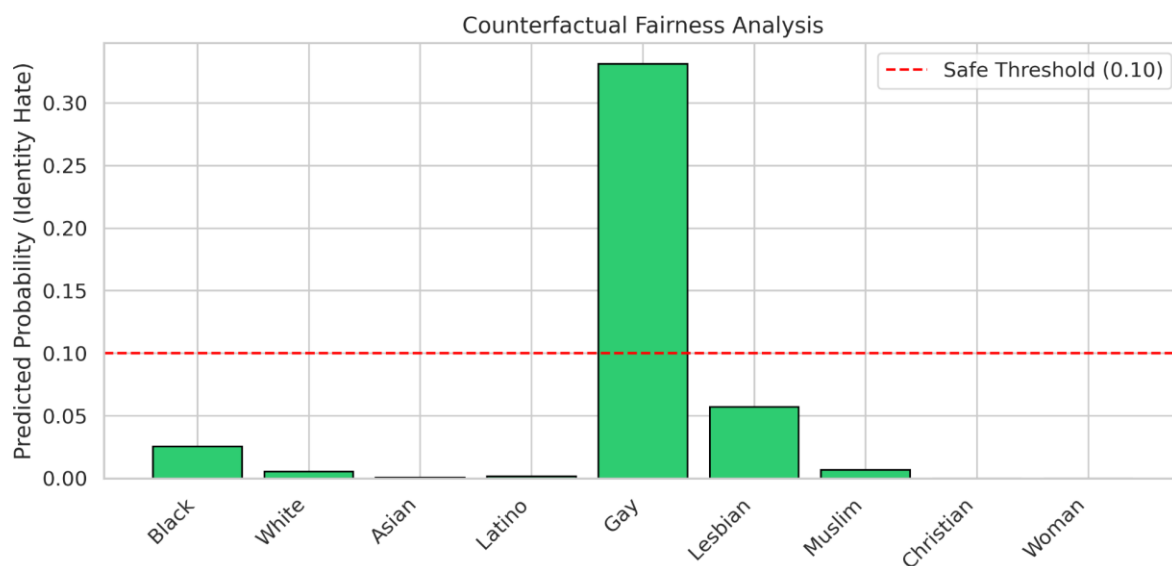


Figure 12. Identity-Term Sensitivity Analysis

While it is not intended to point out whether the terms used by identity labels are harmful or not, the difference in toxicity scores among the different demographics clearly shows the presence of some sort of bias in the data. The present bias evaluation is incomplete in its scope and not a fully developed benchmark of fairness when it comes to demographic distribution.

5.5 Counterfactual analysis

To evaluate the robustness and contextual sensitivity of ToxiFusionNet, counterfactual testing was conducted by substituting semantically critical words and observing the multi-label probability shifts. Small lexical perturbations can drastically alter specific class probabilities without necessarily changing the overall toxicity detection, demonstrating the model's granular sensitivity.

For instance, substituting the word 'stupid' with 'wrong' in the baseline sentence ("You are a stupid idiot and absolutely useless") caused the predicted probability for the obscene class to drop significantly from 0.883 to 0.618 (Table 10). An even more drastic shift occurred when replacing 'idiot' with 'person', which collapsed the obscene probability down to 0.391, effectively dropping it below the classification threshold. Conversely, the toxic and insult probabilities remained highly stable (above 0.930) across these perturbations. This demonstrates that ToxiFusionNet's predictions are highly sensitive to specific semantic triggers for specific classes, rather than relying on spurious, generalized dataset correlations.

Table 10. Counterfactual multi-label probability shifts for the phrase: "You are a [mutation 1] [mutation 2] and absolutely [mutation 3]."

Text Mutation	Toxic Prob.	Obscene Prob.	Insult Prob.
Original baseline	1.000	0.883	0.983
Replace 'stupid' → 'wrong'	0.999	0.618	0.969
Replace 'idiot' → 'person'	0.998	0.391	0.937
Replace 'useless' → 'unhelpful'	1.000	0.876	0.984

The results from the counterfactual evaluation show that ToxiFusionNet does not merely memorize toxic structures but dynamically adjusts its multi-label probabilities based on the exact severity of the lexicon used.

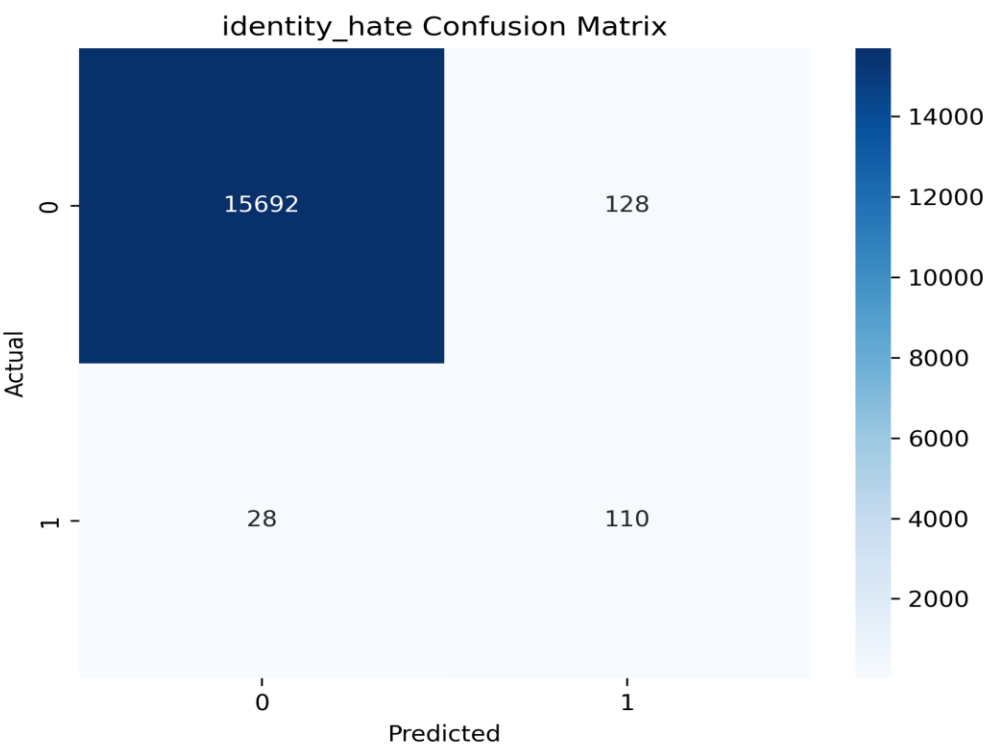
5.6 Error Analysis

To carry out an error analysis for the model, the confusion matrices for each of the toxicity classes were created to analyze their class-level prediction behavior. The confusion matrices provide class-level insight into true positives, true negatives, false positives, and false negatives across each toxicity category. Based on the findings, it can be concluded that the model has high accuracy rates of true negatives in the classes, indicating that it correctly classifies non-toxic data.

As for the cases of high-frequency classes such as toxic, obscene, and insulting, the model exhibits high accuracy rates of true positives, suggesting that it can identify explicit toxicity. For low-frequency classes such as threat and identity hate, the model has high accuracy rates of false negatives, indicating that it cannot identify implicit toxic data. This limitation is likely associated with severe class imbalance and limited minority-class training samples within the dataset.

False positives are also present in the case of borderline instances where it is hard to differentiate between neutral and mildly toxic expressions. Similar challenges involving minority classes, ambiguity, and borderline toxic expressions have also been observed in recent toxicity classification studies [28].

Figure 13. Confusion matrix for the identity hate class showing class-level prediction performance



5.7 Failure analysis

While the system demonstrated strong precision on majority classes, its performance on minority categories (threat and identity hate) was significantly constrained by the limited availability of training samples. Another problem related to the performance of the system was the presence of misclassifications in cases of borderline and implicit toxicity due to ambiguity.

Furthermore, qualitative error inspection revealed that ToxiFusionNet occasionally struggles with deeply contextual sarcasm and intentional adversarial misspellings (e.g., character masking such as 'id!ot'). Because the handcrafted stylistic pipeline partially relies on explicit lexicon matching and standard punctuation distributions, highly implicit toxicity that lacks structural aggression markers can sometimes bypass the fusion layer's detection thresholds. Future iterations of the model must address these adversarial edge cases.

6. CONCLUSION

In this work, we presented ToxiFusionNet, an integrated explainable framework for multi-label toxicity detection that combines transformer-based semantic representations with engineered linguistic and stylistic features. With the help of transformer architecture in combination with contextualized embeddings produced by the BERT model and engineered features, this model can avoid disadvantages of both feature-based and pure deep learning approaches.

Evaluation of the method on the Jigsaw Toxic Comment Classification dataset revealed an advantage of a macro-F1 score of 0.69 compared to the feature-based methods and BERT model as a baseline. Interpretation with the help of SHAP and integrated gradients helped us gain a better understanding of model workings [13, 29]. Furthermore, bias sensitivity evaluation and counterfactual robustness analysis revealed the presence of residual identity-related bias and sensitivity toward adversarial lexical perturbations.

These examples demonstrate that ToxiFusionNet's predicted probabilities respond to targeted lexical substitutions. However, because the analysis is based on a small number of manually constructed perturbations, these observations should be interpreted as qualitative sensitivity analysis rather than as a comprehensive evaluation of robustness.

Indeed, the result shows that the subjectivity of the entire process of the toxicity detection procedure is apparent, and it involves several elements that make this method ineffective due to label noise and human disputes. In such a way, one of the most effective ways of overcoming the problem is the implementation of the ensemble technique along with the threshold optimization for each class.

Nonetheless, some restrictions ought to be noted. The proposed methodology implies the use of extremely heavy transformer models, displays relatively low efficiency for minority classes of toxicity and is sensitive to distributional biases and lexical perturbations. The following areas can be considered for future work: optimization of fairness, multilingual toxicity detection and adversarial learning.

REFERENCES

- [1] Lees, A., Tran, V. Q., Tay, Y., Sorensen, J., Gupta, J., Metzler, D., & Vasserman, L. (2022, August). A new generation of perspective api: Efficient multilingual character-level transformers. In Proceedings of the 28th ACM SIGKDD conference on knowledge discovery and data mining (pp. 3197-3207).
- [2] Dixon, L., Li, J., Sorensen, J., Thain, N., & Vasserman, L. (2018, December). Measuring and mitigating unintended bias in text classification. In Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society (pp. 67-73).
- [3] Ousidhoum, N., Lin, Z., Zhang, H., Song, Y., & Yeung, D. Y. (2019, November). Multilingual and multi-aspect hate speech analysis. In Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP) (pp. 4675-4684).
- [4] Ross, A. S. (2021). Right for the Right Reasons: Training Neural Networks to Be Interpretable, Robust, and Consistent with Expert Knowledge (Doctoral dissertation, Harvard University).
- [5] Pavlopoulos, J., Sorensen, J., Dixon, L., Thain, N., & Androutsopoulos, I. (2020, July). Toxicity detection: Does context really matter?. In Proceedings of the 58th annual meeting of the association for computational linguistics (pp. 4296-4305).
- [6] Tarekegn, A. N., Ullah, M., & Cheikh, F. A. (2024). Deep learning for multi-label learning: A comprehensive survey. arXiv preprint arXiv:2401.16549.
- [7] Zampieri, M., Nakov, P., Rosenthal, S., Atanasova, P., Karadzhov, G., Mubarak, H., ... & Çöltekin, Ç. (2020, December). SemEval-2020 task 12: Multilingual offensive language identification in social media (OffensEval 2020). In Proceedings of the fourteenth workshop on semantic evaluation (pp. 1425-1447).
- [8] Han, X., & Tsvetkov, Y. (2020, November). Fortifying toxic speech detectors against veiled toxicity. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP) (pp. 7732-7739).
- [9] Mozafari, M., Farahbakhsh, R., & Crespi, N. (2020). Hate speech detection and racial bias mitigation in social media based on BERT model. PloS one, 15(8), e0237861.
- [10] Mazari, A. C., Boudoukhani, N., & Djeflal, A. (2024). BERT-based ensemble learning for multi-aspect hate speech detection. Cluster Computing, 27(1), 325-339.
- [11] Di Bonaventura, C., Siciliani, L., Basile, P., Meroño-Peñuela, A., & McGillivray, B. (2025, January). From detection to explanation: effective learning strategies for LLMs in online abusive language research. In Proceedings of the 31st International Conference on Computational Linguistics (pp. 2067-2084).
- [12] Atanasova, P., Simonsen, J. G., Lioma, C., & Augenstein, I. (2020, November). A diagnostic study of explainability techniques for text classification. In Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP) (pp. 3256-3274).

- [13] Sundararajan, M., Taly, A., & Yan, Q. (2017, July). Axiomatic attribution for deep networks. In International conference on machine learning (pp. 3319-3328). PMLR.
- [14] Kurita, K., Belova, A., & Anastasopoulos, A. (2019). Towards robust toxic content classification. arXiv preprint arXiv:1912.06872.
- [15] Hartvigsen, T., Gabriel, S., Palangi, H., Sap, M., Ray, D., & Kamar, E. (2022, May). Toxigen: A large-scale machine-generated dataset for adversarial and implicit hate speech detection. In Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers) (pp. 3309-3326).
- [16] Wu, W., Arendt, D., & Volkova, S. (2021, April). Evaluating neural model robustness for machine comprehension. In Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume (pp. 2470-2481).
- [17] Vidgen, B., Thrush, T., Talat, Z., & Kiela, D. (2021, August). Learning from the worst: Dynamically generated datasets to improve online hate detection. In Proceedings of the 59th annual meeting of the Association for Computational Linguistics and the 11th international joint conference on natural language processing (volume 1: long papers) (pp. 1667-1682).
- [18] Blodgett, S. L., Barocas, S., Daumé Iii, H., & Wallach, H. (2020, July). Language (technology) is power: A critical survey of “bias” in NLP. In Proceedings of the 58th annual meeting of the association for computational linguistics (pp. 5454-5476).
- [19] Kennedy, C. J., Bacon, G., Sahn, A., & von Vacano, C. (2020). Constructing interval variables via faceted rasch measurement and multitask deep learning: a hate speech application. arXiv preprint arXiv:2009.10277.
- [20] Mathew, B., Saha, P., Yimam, S. M., Biemann, C., Goyal, P., & Mukherjee, A. (2021, May). Hatexplain: A benchmark dataset for explainable hate speech detection. In Proceedings of the AAAI conference on artificial intelligence (Vol. 35, No. 17, pp. 14867-14875).
- [21] Jahan, M. S., & Oussalah, M. (2023). A systematic review of hate speech automatic detection using natural language processing. *Neurocomputing*, 546, 126232.
- [22] Röttger, P., Vidgen, B., Nguyen, D., Talat, Z., Margetts, H., & Pierrehumbert, J. (2021, August). HateCheck: Functional tests for hate speech detection models. In Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers) (pp. 41-58).
- [23] Zaheri, S., Leath, J., & Stroud, D. (2020). Toxic comment classification. *SMU Data Science Review*, 3(1), 13.
- [24] Ribeiro, M. T., Singh, S., & Guestrin, C. (2016, August). " Why should i trust you?" Explaining the predictions of any classifier. In Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining (pp. 1135-1144).

- [25] Chen, Z., Zhang, J. M., Hort, M., Harman, M., & Sarro, F. (2024). Fairness testing: A comprehensive survey and analysis of trends. *ACM Transactions on Software Engineering and Methodology*, 33(5), 1-59.
- [26] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019, June). Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies*, volume 1 (long and short papers) (pp. 4171-4186).
- [27] Kaushik, D., Hovy, E., & Lipton, Z. C. (2019). Learning the difference that makes a difference with counterfactually-augmented data. *arXiv preprint arXiv:1909.12434*.
- [28] Sheng, E., Chang, K. W., Natarajan, P., & Peng, N. (2021, August). Societal biases in language generation: Progress and challenges. In *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: long papers)* (pp. 4275-4293).
- [29] Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30.

Data availability statement

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